

Review

Improving the accuracy of global ECMWF wave height forecasts with machine learning

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ABSTRACT

Significant wave height (SWH) stands as one of the most crucial parameters for maritime activities. However, even the SWH data from the widely utilized European Centre for Medium-Range Weather Forecast Integrated Forecasting System (ECMWF-IFS) carries errors and uncertainties. In this study, the Light Gradient Boosting Machine (LightGBM) is used to inference the global ECMWF-IFS SWH forecast biases. The results demonstrate that globally, the LightGBM reduces the root mean square error by 10–20 %. Particularly noteworthy is the enhanced forecast accuracy observed in the western Pacific during late summers. Furthermore, the corrected forecast results during Super Typhoon Lekima in 2019 showcase the capability of model to effectively enhance the forecast accuracy of typhoon-induced wind waves, even when four typhoons occur concurrently. This study establishes the feasibility of LightGBM in inferencing single-step SWH forecast bias and presents a cost-effective model for enhancing global wave forecasts.

1. Introduction

Ocean surface gravity waves, hereinafter, waves, are one of the most important geophysical phenomena with strong nonlinear and nonstationary characteristics. Crucial for climate change and projections, nearshore, coastal, and offshore engineering, maritime transportation, mariculture, and renewable energy (Chen et al., 2021), wave forecasts have been traditionally achieved using a variety of statistical (Bethel et al., 2021) and numerical modeling techniques. Particularly, current numerical techniques are dependent upon the usage of third-generation wave models such as WaveWatch III (Tolman and others, 2009), Wave Modeling (Group, 1988), and Simulating Waves Nearshore (Booij et al., 1999), amongst others. These are based on solving the spectral wave action balance equation through discrete calculations on a grid, instead of differential calculations. This process inevitably introduces some errors (Campos et al., 2018, 2020), such as truncation errors. In the high-resolution numerical prediction of coastal and nearshore waves, numerical models are highly sensitive to the accuracy and inherent

uncertainty of input wind fields, bathymetric data, and applied boundary conditions, all of which further affect the accuracy of model forecasts. To minimize errors and reduce forecast biases, the development, application, and tuning of postprocessing tools can be used to improve the forecast accuracy of numerical wave models (Arribas et al., 2011; Glahn and Lowry, 1972; Stockdale et al., 1998).

However, due to the diverse and stochastic nature of forecast biases, it is difficult for statistical methods to accurately capture the spatio-temporal patterns of said biases. In recent years, artificial intelligence (AI) has been increasingly used in various fields but especially the marine sciences due to its good adaptiveness and strong ability to solve nonlinear problems, such as the automatic detection and forecasting of mesoscale eddies (Xu et al., 2019; Zeng et al., 2015), El Niño forecasting (Ham et al., 2019; Liang et al., 2021; Zhou and Zhang, 2022), Arctic sea ice concentration forecasting (Kim et al., 2018; 2020), and sea surface temperature forecasting (Zhang et al., 2017; Zheng et al., 2020), which also provides new solutions for model forecast bias corrections. Machine learning-based forecast post-processing techniques are more widely

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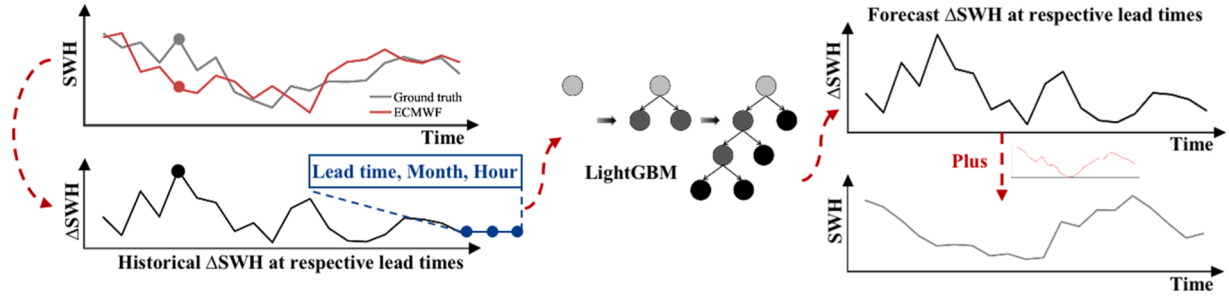


Fig. 1. Schematic diagrams of the ECMWF-LightGBM model.

used in the field of weather forecasting (Bora et al., 2023), such as forecast correction of sea surface temperature (Fei et al., 2022), the Madden-Julian Oscillation (Kim et al., 2021), ECMWF systematic model error in the stratosphere (Bonavita and Laloyaux, 2020), weather variables from ECMWF-IFS: 2 m temperature, 2 m relative humidity, 10 m wind speed, and 10 m wind direction (Han et al., 2021) and other variables.

Previous research has shown that it is feasible to use AI to correct wave forecast errors. Makarynsky (2004) used an Artificial Neural Network (ANN) to forecast the future 24 h wave parameters and correct the initial forecasts to further improve the forecast accuracy. Makarynsky & others (2005) proposed to use of the ANN method to learn and analyze the differences between simulated and measured data of the WAM model for data assimilation of wave parameters such as wave height and wave period. Londhe et al. (2016) used Indian National Centre for Ocean Information Services model wave height historical forecast error as input to revise the wave forecast data for the 24th hour, but the paper only discusses the correction effect of the 24 h forecast and does not discuss the correction effect of other forecast timings. Ellenson et al. (2020) used bagged decision trees, using the output of WW3 and wind speed data to detect forecast bias in wave height, and the forecast root mean square error (RMSE), bias, and percentage error were reduced in several different experiments. Gracia et al. (2021) combined the multilayer perceptron and LightGBM to reduce the numerical model SWH, wave period, and other wave parameters of forecast errors were

reduced by 36 % on average. Sun et al. (2022) innovatively used BU-net to correct the two-dimensional significant wave height (SWH) forecasts of 24, 48, and 72 h, which significantly reduced the short-term forecast errors. However, the single forecast step forecasting model increases the training cost and complexity of use. The preceding research primarily concentrates on correcting short-term SWH forecasts (lead time < 72 h) for AI models or regional numerical models, but there is a scarcity of relevant studies pertaining to the correction of global medium-term SWH forecasts. Such corrections are crucial for ensuring ocean shipping safety and facilitating route planning. The widely utilized ECMWF forecast product is associated with significant forecast errors in its medium-term SWH forecasts owing to various factors such as forcing fields. To address this, the ECMWF-LightGBM is herein proposed to study the predictability of the SWH forecast bias.

The rest of the paper is structured as follows: the data and method are introduced in Section 2, Section 3 evaluates and discusses the corrected SWH forecast. Finally, a summary and discussion is given in Section 4.

2. Data and method

2.1. ECMWF-IFS

In this paper, the numerical weather significant wave height (SWH) forecasting product, publicly released by the ECMWF-IFS (hereinafter, ECMWF) is used. The forecast product is released twice a day, with the

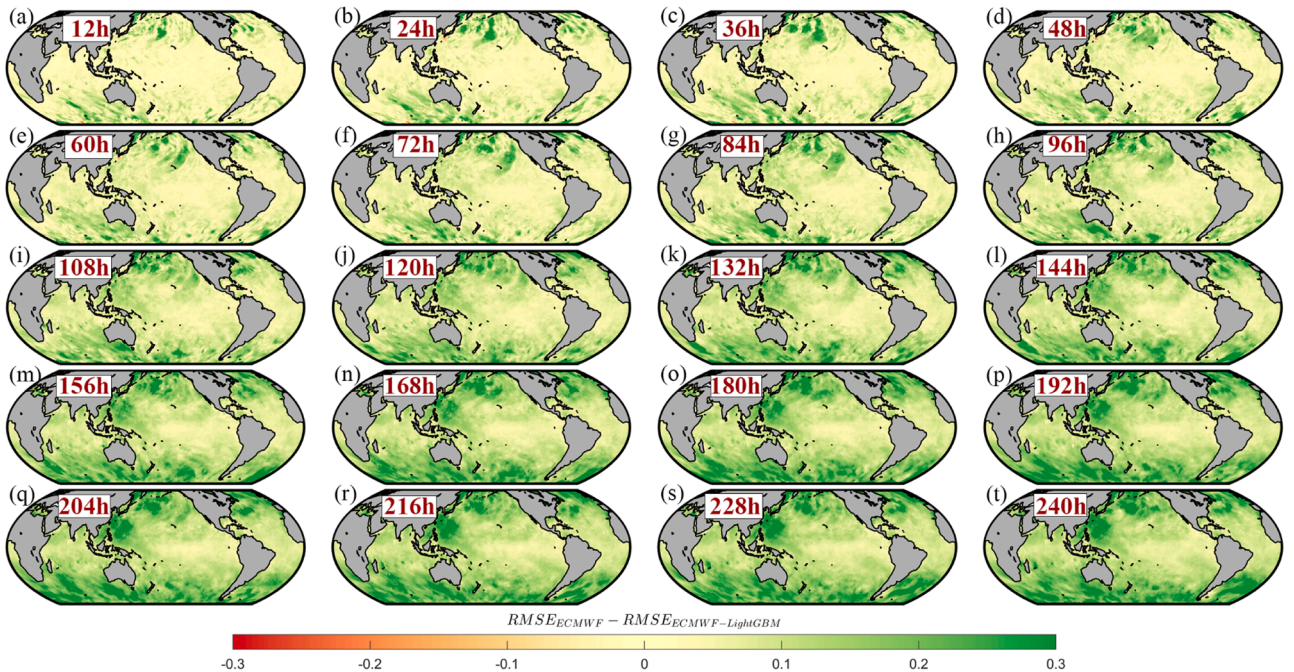


Fig. 2. Spatial distribution of RMSE reduction in SWH forecasts for different lead times after correction, (a-t) for lead times of 12 – 240 h (12 h interval).

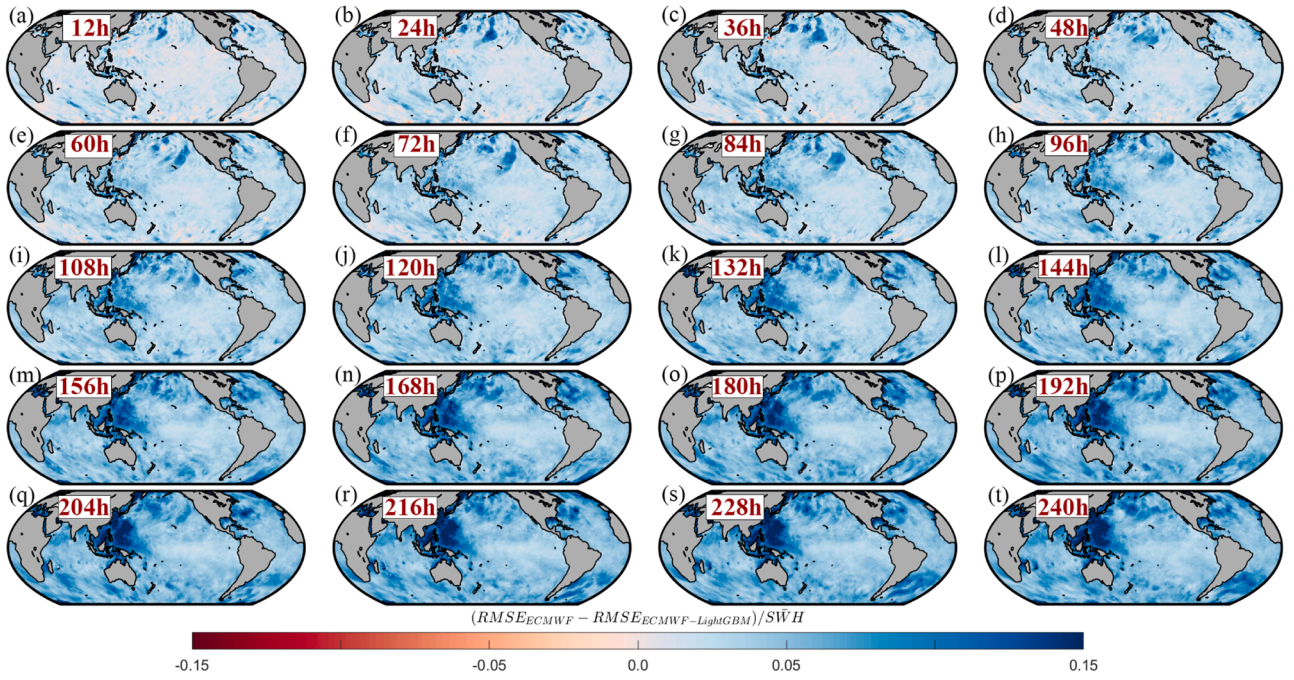


Fig. 3. Spatial distribution of relative reduction rate of RMSE for SWH with different lead times after correction, (a-t) for lead times of 12–240 h (12 h interval).

starting time at 00:00 and 12:00 UTC, and the lead time is 6–240 h with a temporal resolution of 6 h. The spatial resolution of the SWH data is $1/2^\circ \times 1/2^\circ$. The time range of data used in this paper is 2018–2019, due to some data missed, with a total of 624 forecast data for 2018 and 661 forecast data for 2019, using the data for 2018 as the training set and the data for 2019 as the test set, where the missing values are filled using spline interpolation. Since there are more missing data in high-latitude sea areas affected by sea ice, only the SWH in the range of $60^\circ\text{S} - 60^\circ\text{N}$ is used in this paper.

2.2. ECMWF-LightGBM model

Originally proposed by Microsoft as an open-source framework, the LightGBM uses a gradient-boosting decision tree algorithm to achieve more accurate predictions and is a decision tree-based algorithm that divides the parameters in the input layer into different parts to construct a mapping relationship between inputs and outputs. The model uses a

leaf-wise tree growth strategy to speed up the training instead of using a level-wise tree strategy. Leaf-wise trees can significantly reduce the number of tree nodes, and thus can significantly speed up the training process when the dataset is large. Fig. 1 shows the structure of the ECMWF-LightGBM. To facilitate the use in practical operations, the analysis data at time 0 is used as the true value in the model in this paper, and the forecast SWH is corrected for lead times from 12 to 240 h (with 12 h intervals). The inputs to the model are the error series of the historical forecasts at a certain lead time, month, and hour, and the innovative introduction of month, time, and lead time as inputs allows the number of models to be refined. After sensitivity experiments, the length of the error series is 5. The output is the SWH forecast error for a forecast length to be corrected, and the corrected forecast is obtained by adding it to the forecast value to be corrected.

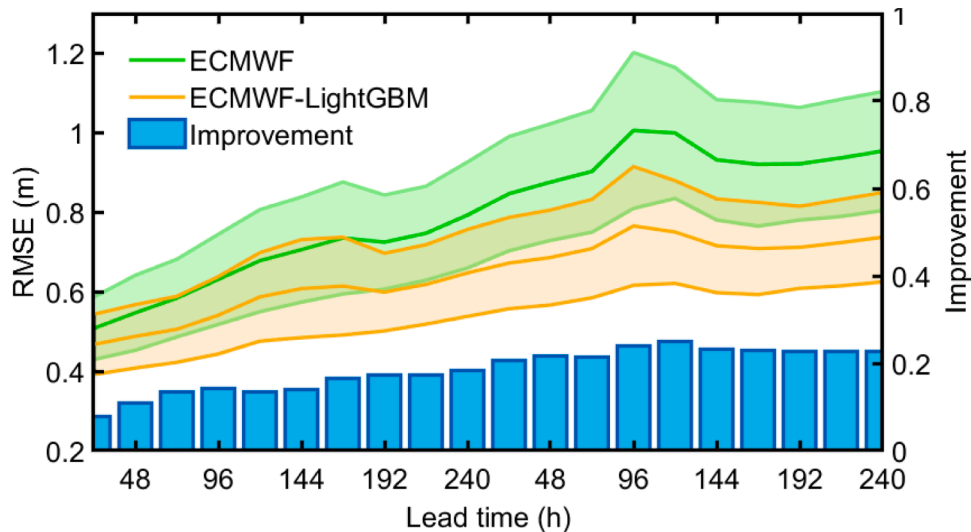


Fig. 4. Variation of RMSE with a lead time for ECMWF forecasts and corrected forecast results.

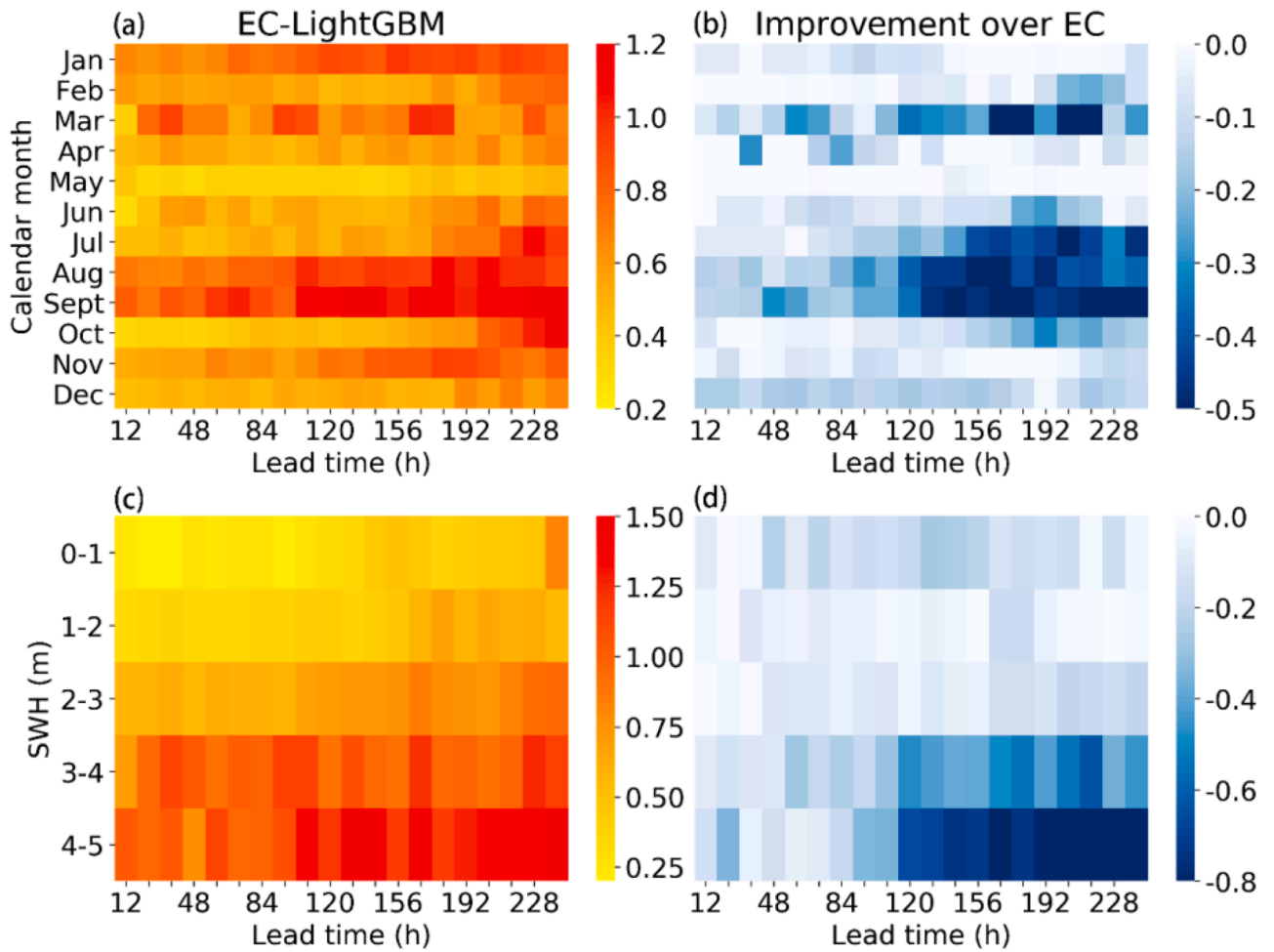


Fig. 5. Heatmaps of ECMWF-LightGBM forecast RMSE with (a) lead time for each month, (b) reduced values relative to the original forecast, (c) lead time for different real SWH, and (d) reduced values relative to the original forecast.

3. Results

3.1. Correction effects of global SWH forecast

Fig. 2 illustrates the distribution of changes in RMSE for SWH forecasts before and after the correction of the ECMWF forecast product in the test dataset. The impact of the correction is most notable in the northern and western Pacific Ocean, the northern Atlantic Ocean, and the Southern Ocean. These regions exhibit higher SWH variability and less observational data for assimilation, posing greater challenges for numerical models to make accurate predictions. Consequently, the magnitude of error correction in these areas is naturally higher than in others. Moreover, the correction has a more pronounced effect on longer lead times, as numerical models deviate further from observations due to imperfect initial conditions, model physics, and inherent limitations in computing resources. In short-term forecasts, apart from the northern Pacific and the Southern Ocean, reductions in RMSE are marginal, and in some areas, such as the southern sea area of Japan, RMSE may even increase. However, in medium-term forecasts, forecast accuracy notably improves after correction, particularly in the northern and western sea areas of the Pacific Ocean.

Due to significant variations in SWH across different ocean basins, assessing the improvement of SWH forecasts solely through RMSE changes presents challenges. To address this, the relative change in RMSE can be calculated by dividing the RMSE change value by the average SWH in 2019 (Fig. 3). This approach provides clearer insights. Notably, for short-term forecast corrections, the effects are less

pronounced, underscoring the effectiveness of current numerical wave models. However, as the lead time increases, correction effects become more noticeable, exceeding 15 % in the western Pacific Ocean north of Australia and east of Indonesia. Although the RMSE change value is substantial, the high average SWH in the Southern Ocean region results in a relatively small relative change.

3.2. Correction effects of northwest pacific SWH forecast

Figs. 2 and 3 show that the impact of SWH correction is most prominent in the western Pacific Ocean. To further investigate the corrective influence of applying LightGBM to SWH forecasts, we focus on the geographic area within 20 – 28°N, 118 – 123°E. In Fig. 4, The SWH forecast RMSE increases as the lead time increases, averaging RMSE of 0.51 m for the 12 h forecast and 0.95 m for the 240 h forecast in the uncorrected results. After correction, the forecast RMSE for 12 and 240 h becomes 0.44 m and 0.71 m respectively. The blue histogram displays the reduction rate of RMSE, revealing a decrease of approximately 10–20 % after the LightGBM correction. As lead time extends, the corrective effects of LightGBM intensify, while the RMSE trend remains consistent before and after corrections. This suggests that the corrections are still heavily influenced by the original data.

Fig. 5a shows that the forecast RMSE is the largest in August – September and increases with lead time, and the RMSE can be reduced by >0.3 m (40 %) for 120 – 240 h forecast. The frequent occurrence of typhoon events in August – September indicates that the corrected model can effectively improve the wave height forecasts under extreme

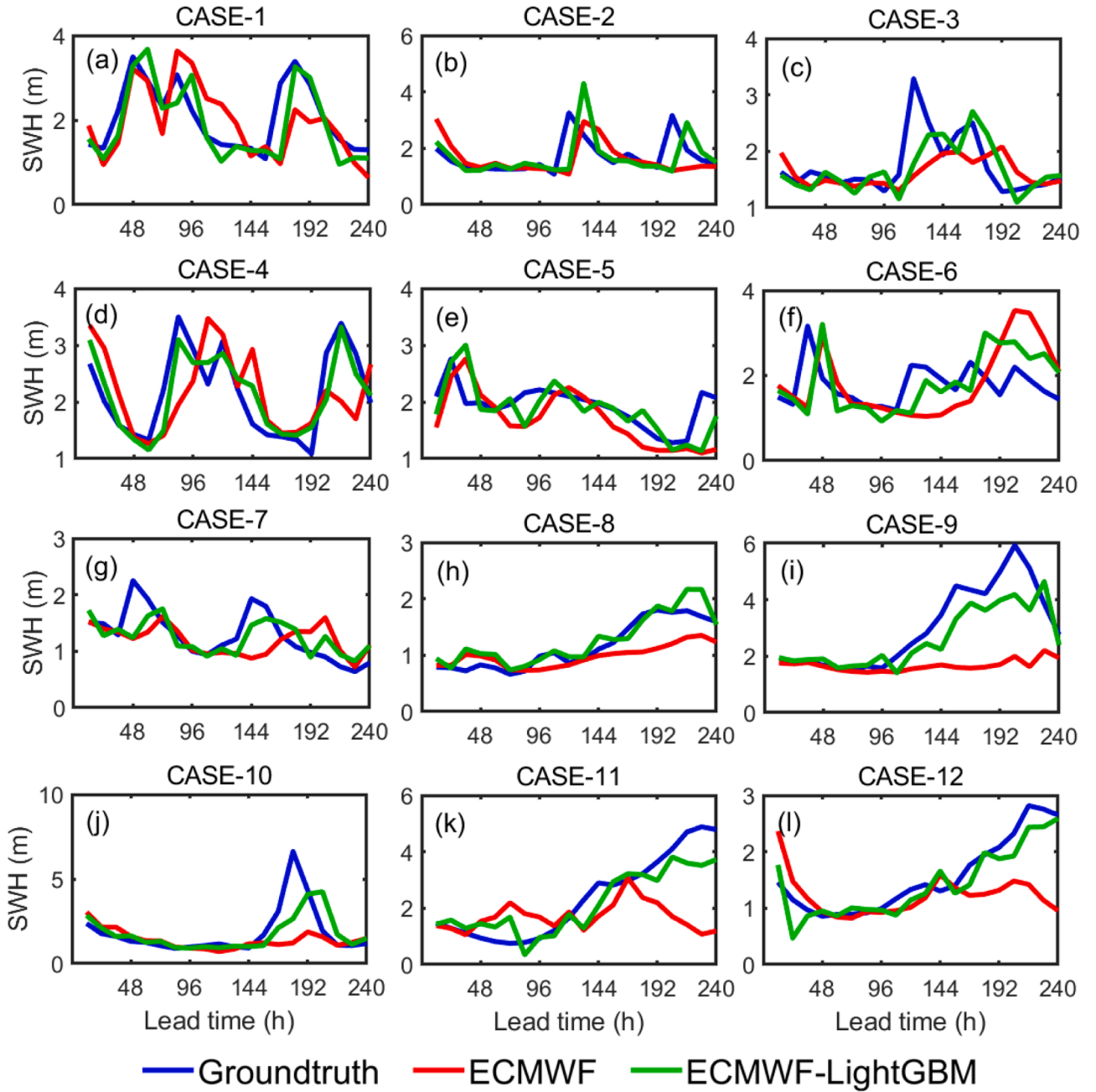


Fig. 6. Evolution of ECMWF-LightGBM single-point SWH corrections over time.

sea states and make up for the shortcomings of the original forecast results in the extreme sea state with large medium-term forecast errors. From Fig. 5c, the RMSE of the ECMWF-LightGB is also related to the true SWH, and the higher the true SWH is, the larger RMSE is. The dark blue area in Fig. 5d is mainly concentrated in the interval of 3 – 5 m waves and long lead time, further suggesting that the ECMWF-LightGBM model can effectively improve the medium-term forecast accuracy of ECMWF forecast products for high SWH.

To better demonstrate the actual correction effect of LightGBM, Fig. 6 shows 12 examples of correction SWH forecasts at position 122.5°E, 23°N in the test dataset. It can be seen that when the ECMWF forecasts deviate from the actual results, the LightGBM can make them more closely match the real values. From the corrected results of Cases 8–12, the ECMWF forecasts are sometimes inaccurate when the SWH increases suddenly, and even the change of SWH cannot be predicted in some cases. However, after the correction, the sudden increase in SWH

can be well predicted. Although the EMD-LSTM method used by Zhou et al. (2021) can also significantly improve the response speed in the case of rising waves, the effect is only significant in the short-term forecast. In the short-term forecasts of individual cases, the corrected values do not differ much from the pre-correction values, while the accuracy of the corrected results in the medium-term is more significantly improved compared to the original forecasts. The average RMSE of the original forecast series is 0.78 m with a correlation of 0.42, and the RMSE after the correction is 0.64 m with a correlation of 0.60, leading to improvements of 18.0 % and 42.9 %, respectively.

3.3. Correction effects of SWH forecast in extreme sea condition

The improvement of prediction accuracy under average statistics does not mean that the SWH prediction skills under extreme sea conditions will be improved. Therefore, we specifically selected the data

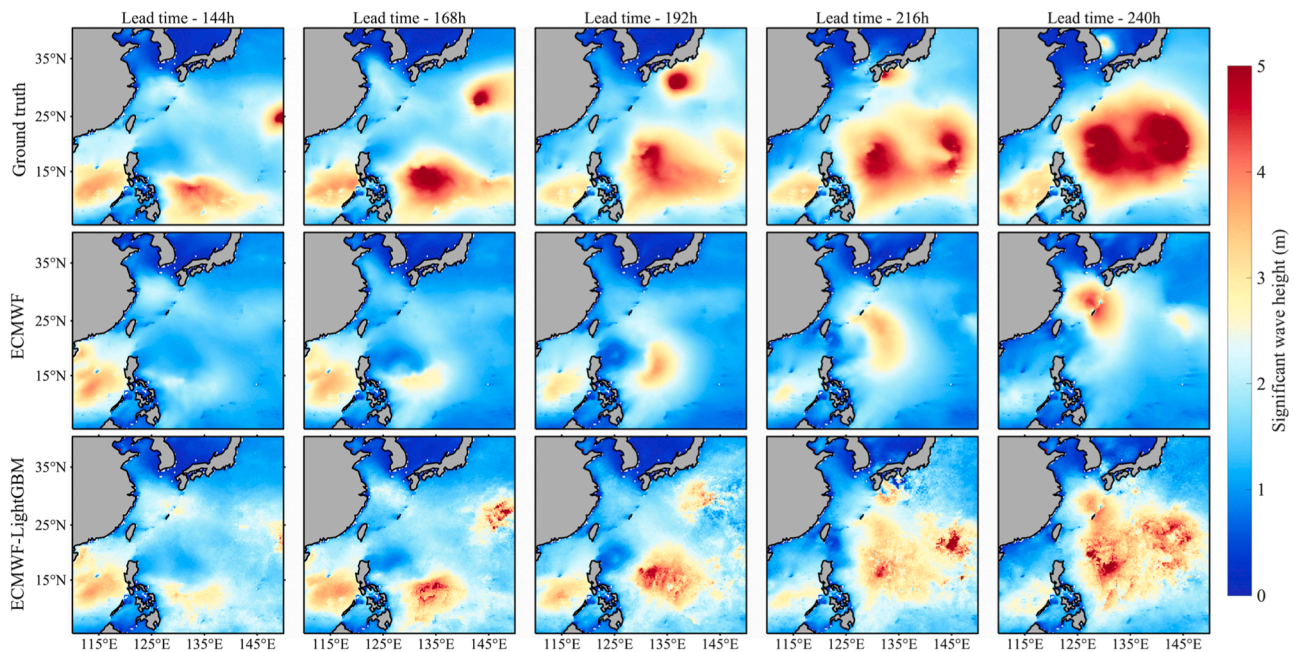


Fig. 7. The SWH of Ground truth (first row), ECMWF (second row), and ECMWF-LightGBM (third row) during super typhoon Lekima in 2019 at lead time from 144 to 240 h (20,190,804–20,190,812). The forecast starting time is 00:00 on July 29, 2019.

during super Typhoon Lekima in 2019 to further evaluate the correction effect of LightGBM. In addition to Lekima, Typhoons Wipha, Francisco and Krosa also jointly affected the northwest Pacific during this period. Fig. 7 shows the forecast results from 144 to 240 h (20,190,804–20,190,812) when the start time is 00:00 on July 29, 2019. At the 144 h lead time, Wipha, had already made landfall, and Lekima and Francisco had just grown into tropical storms. The ECMWF does not forecast the typhoon-induced wind waves caused by Lekima and Francisco, while ECMWF-LightGBM could roughly forecast the high-value center of waves. And the ECMWF did not forecast the typhoon-induced wind wave caused by Lekima until 168 h lead time. At the 216 h lead time, four typhoons were churning the northwest Pacific at the same time, but the ECMWF only shows two high-value centers caused by typhoons. The centers of four typhoon-induced wind waves could be clearly seen after correction. Meanwhile, the point-by-point correction strategy leads to non-smoothness of the corrected results. The above proves that ECMWF-LightGBM can correct the SWH forecast bias under extreme sea conditions.

4. Summary and discussion

In summary, the LightGBM-based model proves the predictability of SWH forecast bias, especially under extreme sea conditions and long lead time. After the correction, the original forecast RMSE can be reduced by about 10 – 20 %, and the correction effect is especially significant in the western Pacific and non-significant near the equator. Our approach has proven effective in improving the SWH prediction skills of ECMWF even in extreme sea conditions with four typhoons at the same time. The corrected-result responds faster to the sudden rise of waves (Fig. 6). However, this study only infers the single-step forecast bias of every lead time, and predictors needs to be adjusted for actual service when applied. The study on the predictability of its biases, suggest that adding historical forecast bias as a predictor can improve the accuracy of correction with long lead time. Because it may not be realistic to directly forecast the bias sequence over a long time, this requires further study in the future. Admittedly, numerical wave simulation forecasting has been fully developed in the past decades, but there is still some room for improvement in forecast accuracy due to the inherent shortcomings of numerical models. More accurate medium-term wave forecasting can

provide stronger guarantees for the route setting and navigation safety of ocean-going vessels, which is important for coastal tourism, aquaculture, and offshore engineering. Meanwhile, accurate wave forecasting helps to calculate the momentum and heat exchange at the air-sea interface more precisely.

Another significance of this study is that the neural network-based correction model can be called directly on a private computer without the need for a mainframe computer, which can be easily installed on a shipboard computer for use. The model is simple to be executed, requiring only few data, effectively reducing the amount of data transfer. In addition, the model is highly flexible, and when applied to the SWH corrections at fixed location sites, such as offshore rigs and islands, more site observations can be added as forecast factors to further improve the correction performance.

Although the SWH forecast products can be improved by machine learning, the improvement of forecast accuracy is still limited. Machine learning is like a black box, which cannot explain what causes the forecast errors and may violate the law of conservation for long lead time forecast. Interpretable and physics-informed neural networks possibly answer these questions, which will be our focus in the future. It is hoped that by combining interpretable or physics-informed neural networks with numerical models, we can gain more insight into the causes of model forecast bias and further improve and enhance the accuracy of model forecasts.

CRediT authorship contribution statement

Shuyi Zhou: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Jiuke Wang:** Data curation. **Yuhan Cao:** Methodology. **Brandon J. Bethel:** Writing – review & editing. **Wenhong Xie:** Software, Methodology, Conceptualization. **Guangjun Xu:** Visualization. **Wenjin Sun:** Writing – review & editing. **Yang Yu:** Data curation. **Hongchun Zhang:** Writing – review & editing. **Changming Dong:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and data availability

ECMWF forecast data can be obtained from <https://data.ecmwf.int>.

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